ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

Tutorial questions

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Topic 1: Pass-Level Tutorial Questions

1. An analogue signal contains frequencies up to 10 kHz. Suppose that we sample this signal at a sampling frequency f_s = 8kHz. What happens to the frequency f_1 = 5 kHz? (Tick the correct answer)
   a) 5kHz frequency will remain unchanged
   b) 5kHz will alias to 3kHz
   c) 5kHz will alias to 0Hz
   d) None of the above

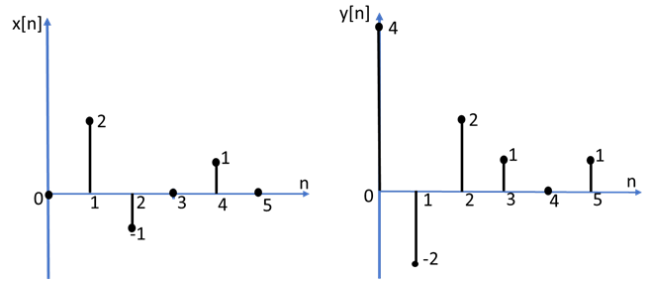
2. Which of the following is not a periodic signal ? (Tick the correct answer)
   a) x[n] = sin[πn/9]
   b) x[n] = sin[nπ]
   c) x[n] = cos[πn^2/$\sqrt{15}$]
   d) x[n] = sin[πn/5 + π] + cos[πn/10 – π]

3. Given a real-valued bandpass signal x_a(t) whose spectrum of X_a(f) nonzero only for $9\ kHz < |f| < 12kHz$, determine the minimum sampling frequency , f_s , for which no aliasing occurs? (Tick the correct answer)
   a) f_S = 18 kHz
   b) f_S = 6 kHz
   c) f_S = 24 kHz
   d) None of the above

4. A discrete-time signal $x[n]$ = 6.35 cos(nπ/10) is quantised with a resolution Δv = 0.2 $volts$. Assuming ADC covers the full signal range , how many bits are required? (Tick the correct answer)
   a) 6 bits
   b) 8 bits
   c) 12 bits

5. Consider the amplitude modulated signal $x(t) = 5\,cos(3\pi t)\,cos(\pi t)$. Determine the frequency components [you may assume that the frequencies are in kHz) (Tick the correct answer)
   a) $\pm$1.5kHz and $\pm$0.5kHz
   b) ± 3kHz and $\pm$1kHz
   c) $\pm$2kHz
   d) $\pm$2kHz and $\pm$1kHz

6. Given the discrete-time sequences $x[n]$ and $y[n]$ shown, which difference equation best describes their relationship ? (Tick the correct answer)



   a) y[n] = x[n-1] + 2x[n+1]
   b) y[n] = 2x[n] – x[n-1]
   c) y[n] = 2x[n-1] + x[n-2]
   d) y[n] = 2x[n-2] – x[n]

7. Determine the period of the discrete time signal $x[n] = \sin(0.2n + \pi)$. (Tick the correct answer)
   (a) 1 sample (b) 5 samples
   (c) 10 π samples (d) Non of the above

Topic 1: Pass-Level Tutorial Questions

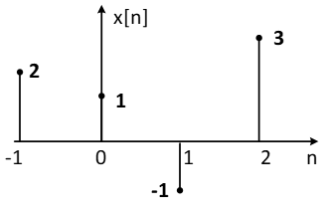
1. A sinusoidal signal is sampled at 50 kHz using a uniform B-bit A/D converter. Assuming that the sinusoid uses the full input range of the A/D converter, determine the minimum number of bits required to achieve an SQNR of at least 92 dB
2. Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by

$$SQNR = 6.02B + 4.77 - 20\log\left(A/\sigma_{sig}\right)$$

where the input range of the A/D converter is $\pm$A and the rms value of the input signal is σ_{sig}. Determine the SQNR if *B* is 16 bits and the input is a signal with a rms value of A/5.

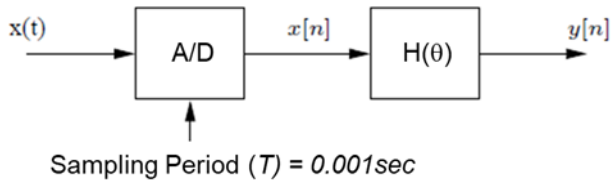
3. An analogue signal $x_a(t) = \cos(2\pi 500t)$ is sampled at a rate of f_s = 4kHz. Determine the resulting analogue and digital magnitude spectra.
4. An analogue signal *x(t)* = $\sin(480\pi t) + 3\sin(720\pi t)$ is sampled 600 times per sec:
   (a) Determine the Nyquist sampling rate for $x(t)$.
   (b) Determine the folding frequency (or half the sampling frequency).
   (c) What are the frequencies, in radians, in the resulting discrete time signal $x[n]$?
   (d) If $x[n]$ is passed through an ideal D/A converter what is the reconstructed signal $y(t)$?
5. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of $\pm$5 Volts is used to convert sampled voltages to integer codes.
   (a) Calculate the quantization interval of the A/D converter.
   (b) Determine the maximum quantization error.

6. Consider the discrete-time sequence x[n] = $\cos[(n\pi/8) + (\pi/4)]$.
   (i) Determine the fundamental period of $x[n]$.
   (ii) If the sampling frequency f_s = 10 kHz, determine one corresponding analogue signal $x(t)$, assuming its frequency is below the Nyquist frequency.
7. Sketch the following:
   (c) x[n] = δ[n+2] + 2 δ[n-1] - 0.5 δ[n-3]
   (d) x[n] = u[-n-2]
   (e) x[n] = 0.5^n u[n-2]
8. Write the equation for x[n] (see diagram) as a sum of scaled and delayed unit impulses



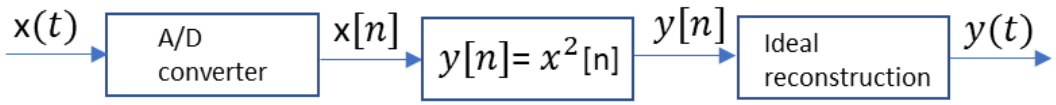
9. For the system shown below , the input x(t) = $\cos(400\pi t) + \cos(600\pi t)$. Find the output y[n] where

$$H(\theta) = \begin{cases} 1 & |\theta \le 0.5\pi| \\ 0 & otherwise \end{cases}$$



Topic 1: Pass-Level Tutorial Questions

10. A speech signal is band limited to 4 kHz is ideally sampled at a sampling rate, f_s. the resulting discrete-time signal $x[n]$, is squared to give $y[n] = x^2[n]$, and $y[n]$ is ideally reconstructed to produce $y(t)$.



What is the minimum sampling frequency f_s, required to ensure $y(t) = x^2(t)$?

11. The signal $x(t) = 2\cos(2000\pi t) - 4\sin(4000\pi t)$ is quantised by rounding, using a 12-bit quantiser with an input range of ±6 V. What is the rms quantisation error and the SQNR?

12. Consider a signal, $s(t)$, with a magnitude spectrum satisfying the following equation

$$|S(f)| = \begin{cases} 3 - |f| & 2\text{kHz} < |f| < 3\text{kHz} \\ 0 & \text{otherwise} \end{cases}$$

f – frequency in kHz. The signal s(t) is sampled uniformly with a sampling frequency of 2kHz. Sketch the magnitude spectrum of the resulting discrete –time signal $|S(\theta)|$ over $-\pi \leq \theta \leq \pi$

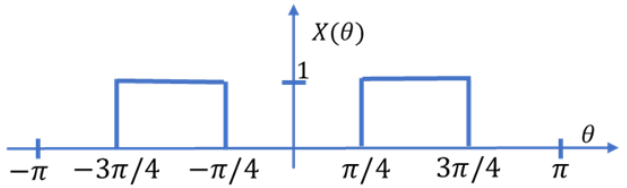
Topic 1: Higher – Level Tutorial Questions

1. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An 8 - bit uniform A/D converter with an input range of ± 5 Volts and rounding to the nearest quantisation level is used to convert sampled voltages to integer codes. At which sampling instant $t = nT > 0$ does the magnitude of the quantisation error first reach its maximum value?
2. A system samples a signal x(t) = $\cos(\omega_o t)\cos(3\omega_o t)$ at f_s = 7 kHz and reconstructs the signal using an ideal LPF with a cutoff frequency of 3.5 kHz. The value of ω_o is $2\pi \times 10^3$ rad.
   (a) What is the highest frequency component present in the signal?
   Hint : $\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$
   (b) Sketch $|X(\omega)|$ and $|X(\theta)|$ – the magnitude spectrum of the signal before and after sampling for (a) above.
   (c) If the sampling frequency is increased to 8 kHz, does aliasing occur? Explain carefully what happens to the 4 kHz component.
3. If an analogue signal *x*(*t*) is sampled every *T* seconds, then at the output of the Analogue to Digital Converter, as shown in a diagram below. Using Inverse Fourier Transforms in Continuous time and discrete time domains, show that

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$



4. Find the Inverse DTFT of $X(\theta)$ for the figure shown below and show that for $n \neq 0$, $x[n] = \frac{2\sin(n\pi/4)\cos(n\pi/2)}{n\pi}$. Also determine $x[0]$.



$$X(\theta) = \sum_{n=-\infty}^{\infty} x(n)e^{-jn\theta}$$

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}d\theta$$

DTFT pair

θ = digital frequency; $-\pi \le \theta \le \pi$

5. Given a real-valued bandpass signal $x_a(t)$ and $X_a(f) = 0$ for $|f| < f_1$ and $|f| > f_2$, the Nyquist sampling theorem says that the minimum sampling frequency is $f_s = 2f_2$. However , in some cases , the signal may be sampled at a lower rate.
   (a) Suppose that f_1 = 8 kHz and f_2 = 10 kHz. Draw a sketch of $X_a(\theta)$, if the sampling frequency f_s = 4 kHz.
   (b) If the bandwidth of the bandpass signal is defined as B = f_2 - f_1 and the centre frequency to be $f_c = \frac{(f_2+f_1)}{2}$ and $f_2 = mB$, where m is a positive integer, show that no aliasing will occur if $x_a(t)$ is sampled at $f_s = 2B$.
   (c) Explain why sampling at $f_s = 2B$ does may cause aliasing when f_2 is not an integer multiple of the bandwidth B.